

Mispricing convertible bonds: a quantitative trading strategy

Fund managers' information

- Rain Condie, Student ID = 12449876
- Vivian Fang, Student ID = 12266409
- Isabel Krogh, Student ID = 12449772
- Justin Wong, Student ID = 12243936
- Yan Zhuang, Student ID = 12459171

Agenda

Strategy Development

- Identifying & exploiting mispricings
- Investment Universe
- Strategy Visualized

Market Impact & Execution

- Market Impact & Profitability
- CB Pricing Model & Trading Signal

Risk & Hedging Approach

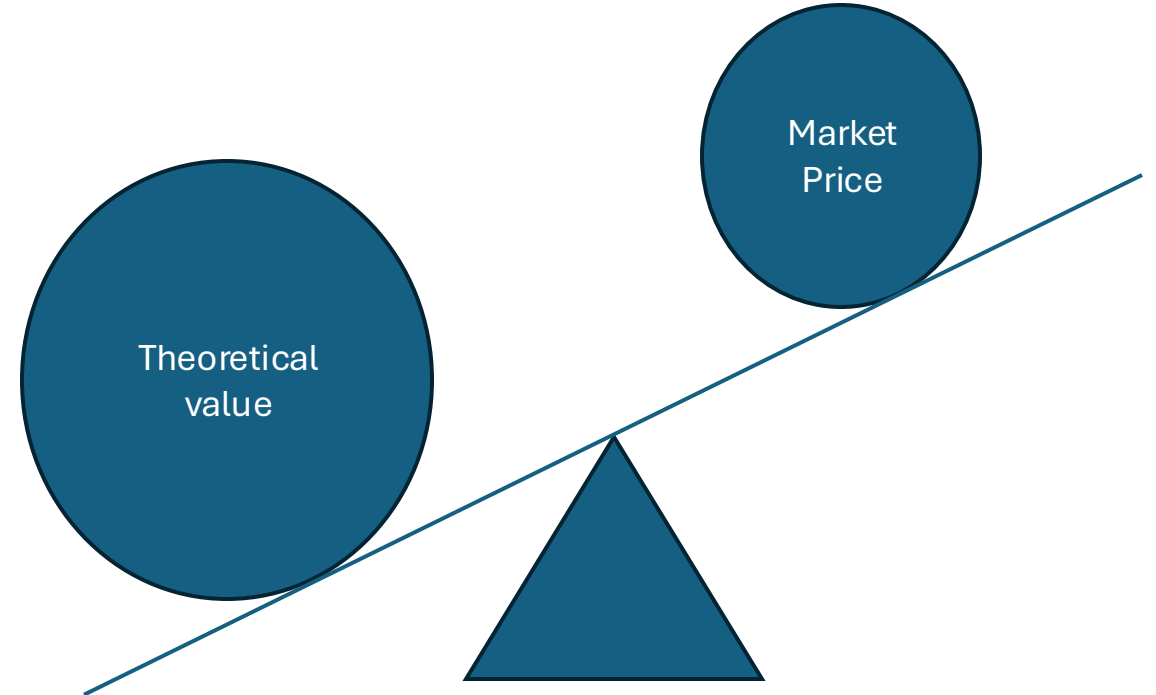
- Position Review
- Hedging Framework
- No Stop-Loss

Performance & Results

- PnL and Sharpe Metrics
- Comparison to Benchmarks

Strategy Overview

- Identify and exploit mispricings in the convertible bond market.
- Buy undervalued convertible bonds and hedge exposure with equity.
- Hedge and manage duration with Treasury securities.
- Profit from price convergence as market value aligns with theoretical value.



Understanding Convertible Bonds

What Are Convertible Bonds?

Hybrid securities that pay interest like bonds but can convert into stock, offering equity upside.

Key Features:

- **Conversion Price & Ratio:** Determines when and how many shares bondholders receive.
- **Coupon & Maturity:** Fixed interest payments with bond-like characteristics.

Why They Matter:

- **For Firms:** Lower borrowing costs, delayed dilution, and strategic flexibility.
- **For Investors:** Downside protection, equity upside, and arbitrage opportunities.
- Convertible bonds combine debt stability with stock potential, making them valuable for both companies and traders.

Investment Universe

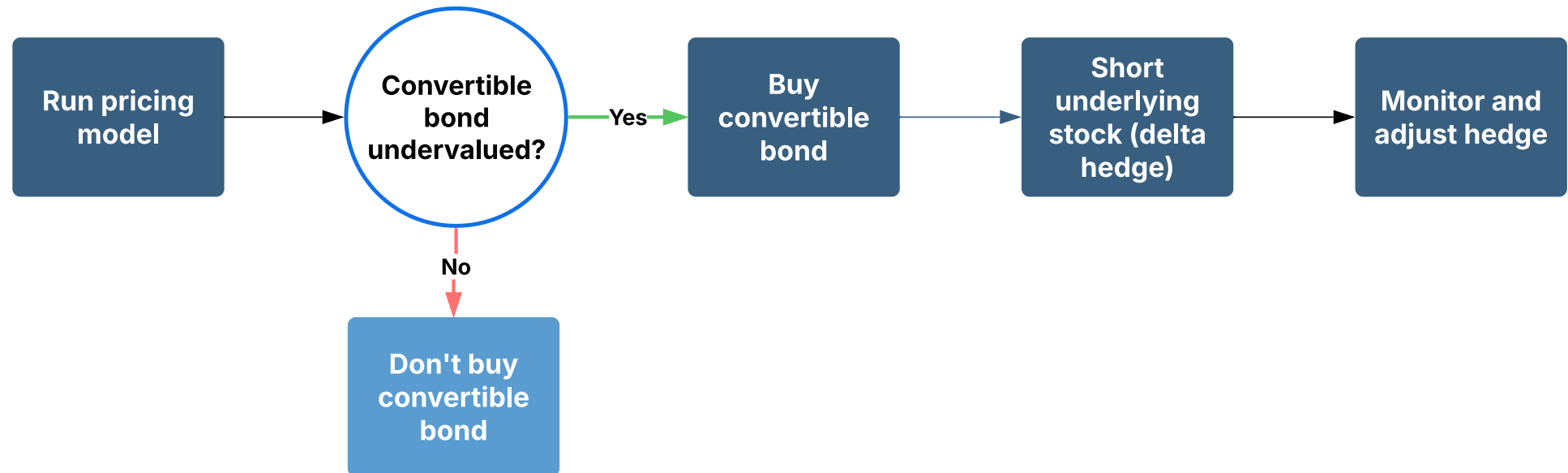
The investment universe for backtesting our trading strategy consists of the **73** most liquid* convertible bonds from each year in **2018 – 2025** that have:

1. **Daily convertible bond prices**
2. **Daily underlying stock prices**
3. **Historical dividend yields for underlying equity**
4. **Bond Characteristics:** conversion price, conversion ratio, coupon rate, maturity date
5. **Credit Spread & Implied Volatility** CDS spread data and implied volatility estimates.
6. **Risk-Free Rates:** U.S. Treasury bond yields

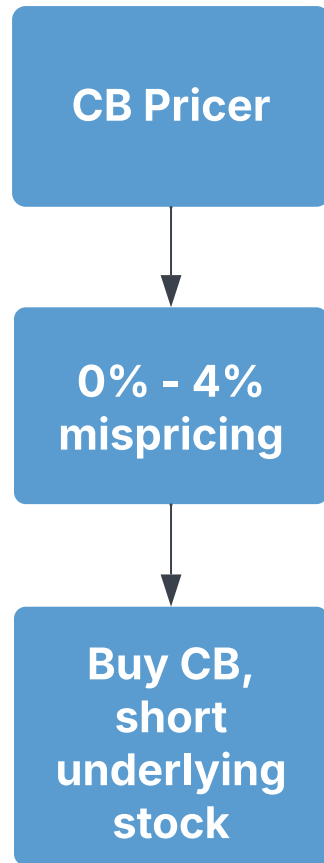
In practice, we would add a new sample of the most liquid convertible bonds each year.

* As defined by Bloomberg's relevance feature

Strategy Visualized



CB Pricing Model & Trading Signal



How We Price a Convertible Bond

We use a **bond + option pricing model**:

- **Bond Component:** Factors in risk-free rate, credit spread, coupon rate, principal, and tenor
- **Option Component:** Uses Black-Scholes to account for stock price, strike, volatility, risk-free rate, and dividends

Trading Signal Based on Mispricing

- If mispricing is **between 0% and 4% (cheap)** --> **Buy CB, short underlying stock**
- If mispricing **exceeds 4%** --> **No trade** (we assume model error).

Market Impact and Profitability

Market Movement	Convertible Bond Value	Short Stock Position	Net Impact
Stock Price Rises	Increases	Loses money	Small loss, can rebalance hedge
Stock Price Drops	Slightly falls (floor value remains)	Gains	Profitable if delta hedge maintained
Bond Price Falls (e.g., credit spread widens)	Loss on bond	Stock hedge may not fully offset	Risk of negative P&L

Why Now? Market Trends & Timing

OTC & Price Inefficiency

- **OTC & Price Inefficiencies** → Convertibles trade over-the-counter with delayed price discovery, creating arbitrage opportunities.

Liquidity Disruptions

- **Liquidity Disruptions** → Forced selling during crises and limited market participants lead to persistent mispricings.

Shift in Arbitrage Strategies

- **Shift in Arbitrage Strategies** → Traditional CB arbitrage is evolving, favoring dynamic hedging—exactly what our strategy targets.

Now is the time to capitalize on these structural inefficiencies with a systematic CB arbitrage approach.

Risk Management: Process & Mitigation

Position Review

- **Maintain, exit, or rebalance** based on mispricing signals
- **Adjust hedges dynamically** (Treasuries & stock positions)

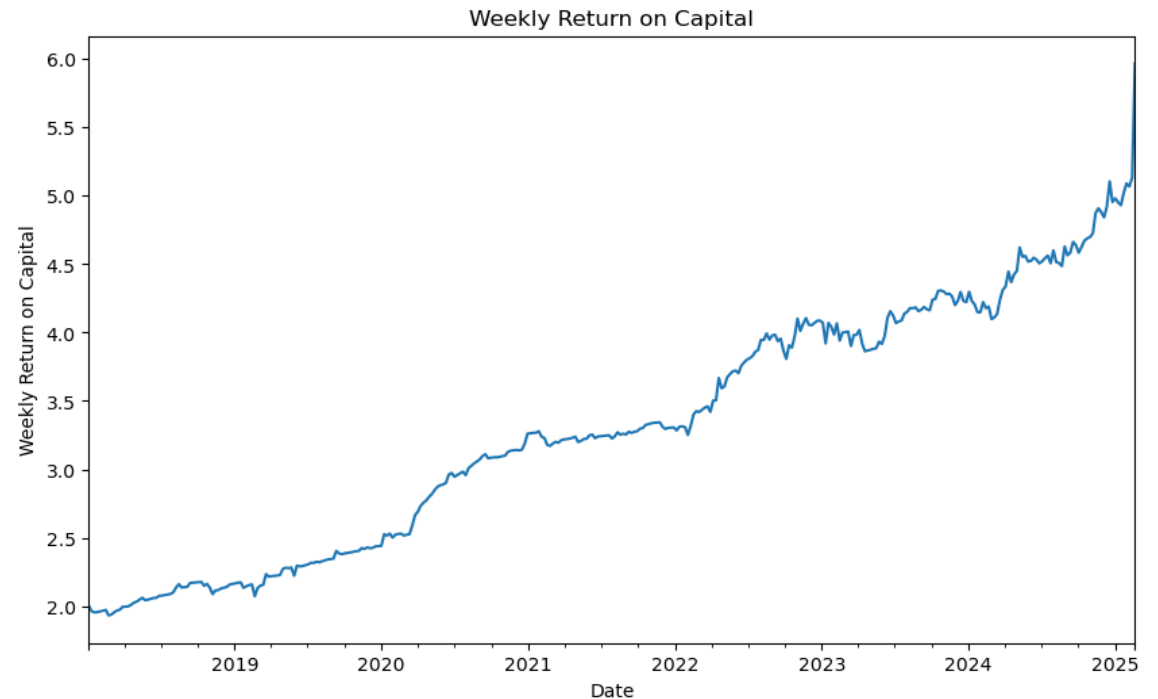
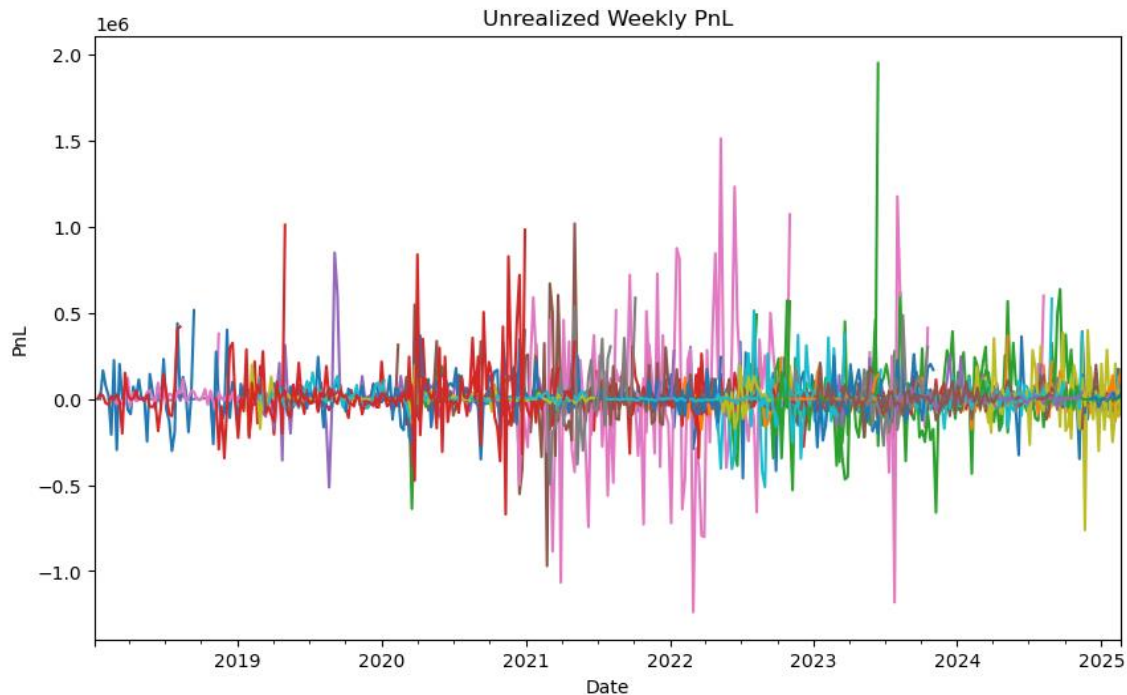
Hedging Framework

- **Duration Risk** → Short **Treasuries** to neutralize interest rate exposure
- **Delta Risk** → Hedge **stock sensitivity** with short positions
- **Ongoing hedge adjustments** to optimize exposure

No Stop-Loss Strategy

- Losses are **managed through dynamic hedging**, not fixed exits
- **Mispricings correct over time**, ensuring profitable mean reversion

Results: PnL, Sharpe, Weekly Returns



- **Annualized Mean Return: 1.9%**
- **Annualized Sharpe Ratio (0.66)** suggests a moderate return per unit of risk.
- **Realized PnL is more stable** with lower volatility compared to **Unrealized PnL**.

Results: Comparison to Market Benchmarks

Low Correlation with Market Indices

- **Realized P&L** shows **minimal correlation** (~ -0.02 to 0.03) with major indices.
- Indicates **market-neutral performance** and **low dependence on equity trends**.

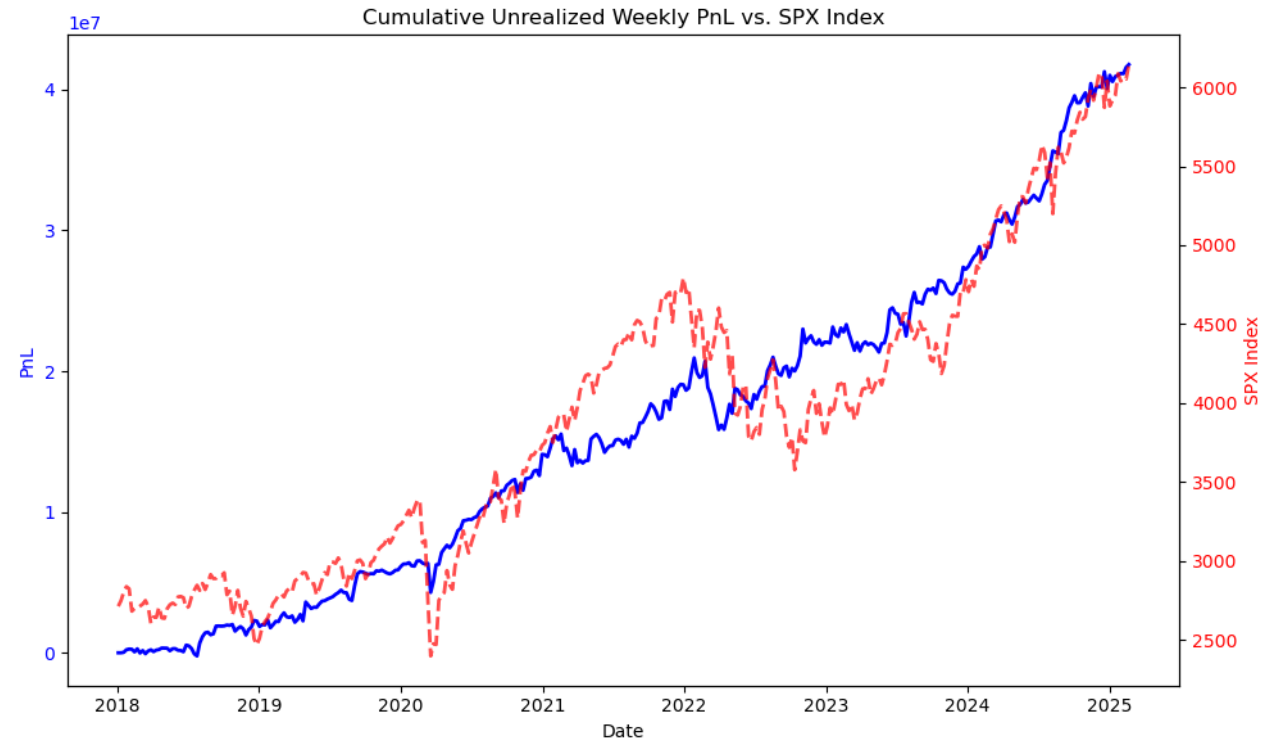
Nasdaq Sensitivity

- **Unrealized P&L** has a slight **negative correlation** (-0.14) with the **CCMP Index**.
- Suggests that **Nasdaq 100 movements** may impact **open CB positions**.

Moderate Link to Credit & Bond Markets

- Indicates that **convertible bond pricing** is **partially influenced by broader market conditions**.

The strategy is **largely independent** of equity market movements, with **some sensitivity to high-yield credit conditions**.



Value-Add of Our Project

Addressing Market Inefficiencies

- Convertible bonds trade **OTC**, leading to:
 - **Limited liquidity** and **manual execution**
 - **Slow price discovery** and **fragmented data**

Our Solution

- **Data Collection & Structuring:** Build a comprehensive dataset of CBs (price history, issue size, volume, conversion terms).
- **Automated Trading Strategy:** Develop a systematic arbitrage model to **identify and execute trades efficiently**.

Results

- **Low Correlation to Benchmarks:** Our strategy exhibits **close to zero correlation** to traditional benchmarks, enhancing diversification.
- **Robust Pricing Framework:** The **bond + option model** effectively prices CBs for **theoretical value**, improving trade precision.

Possible Extensions

Enhancing Credit Assumptions

- Develop a **custom credit model** using **sentiment analysis** and financial statement parsing.
- Leverage **news sentiment, earnings reports, and credit spreads** to refine default risk assumptions.

Improving the Pricing Model

- Upgrade from a **Bond + Option Model** to a more sophisticated **Monte Carlo simulation** or **binomial trees** for improved accuracy.
- Incorporate **early conversion, issuer calls, and other complex option features** to enhance pricing precision.

Advanced Arbitrage Strategy

- Introduce **machine learning** to detect arbitrage opportunities in real-time.
- Implement **dynamic hedging** techniques for optimizing delta, gamma, and vega risk.

References

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